Vishnu Panganamamula . Jarvis Consulting

Hello! I am Vishnu Panganamamula and I am a recent computer engineering graduate (June 2025) from Toronto Metropolitan University (formerly Ryerson University). I have hands-on experience working with data engineering, backend development, cloud and AI/ML engineering. I currently work at Jarvis as a data engineer, where I work on projects that will prepare me for industry level work in the future. I am eager to contribute to meaningful projects where I can use my problem-solving and programming skills to create impactful solutions.

Skills

Proficient: Python, Java, Linux/Bash, RDBMS/SQL, Agile/Scrum, Git

Competent: FastAPI, Docker, Tensorflow, PyTorch, Google Cloud/Azure, HuggingFace, LLMs, sci-kit learn

Familiar: React.js, Node.js, RabbitMQ, Javascript, Kubernetes, C/C++, Firebase, NLTK

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_VishnuPanganamamula

Cluster Monitor [GitHub]: Used Linux commands, Bash scripting, PostgreSQL database and Docker to create a Linux Cluster Monitoring Tool. The monitoring system records specific hardware information of a host VM and what resources of the system are being used. This usage data is recorded and monitored from a host VM running Rocky Linux in real time. The data for the host usage is collected and stored in the PostgreSQL database regularly, allowing users to check their VM usage activity.

Highlighted Projects

Societal Unrest Prediction Model [GitHub]: Built logistic regression and random forest models to predict monthly unrest events across 50 regions (10 years of data), achieving F1/AUC-ROC > 90% Improved recall from <5% to ~95% through feature engineering (temporal + interaction-based) and time-series cross-validation Developed policy analyst alert rules using feature importances and probability thresholds for actionable insights

Medical Imaging Diagnosis Tool [GitHub]: Built and optimized CNN-based models (ResNet-50, DenseNet-121), achieving 90%+ accuracy Enhanced x-ray image interpretability using U-Net segmentation for disease localization Optimized tf.data pipelines to improve training performance, efficiency and prediction capabilities Improved training times from 20 minutes per epoch to 2 minutes per epoch after pipeline optimization Improved model accuracy from 15% to 90%+ for both CNN-based models after pipeline optimization

Professional Experiences

Data Engineer, Jarvis (July 2025-present): Developing industry related projects within a dynamic Agile/Scrum environment.

AI Software Developer Intern, Vosyn Inc. (Sept. 2023-Dec. 2023): Researched and prototyped LLM, TTS and STT systems, shaping architecture for Vosyn's AI platform Fine-tuned Bark TTS, improving multilingual accuracy by 10% across 3 langauges Customized Whisper (LLM with 60M parameters) for Japanese TTS, boosting translation accuracy by 8% Developed scalable backend and database using Django/PostgreSQL, supporting seamless integration Authored risk/safety assessments of for LLM features, identifying 20 bottlenecks and recommending strategies to mitigate risks

Education

Toronto Metropolitan University (Sept. 2020-June 2025), Bachelor of Engineering, Electrical and Computer Engineering - Dean's List (Fall 2024): Earned Dean's List award for Fall 2024 for outstanding academic excellence

Miscellaneous

- Combat Sports (Boxing)
- Casual Gamer
- Drummer